

DATA QUALITY REPORT

Dataset: Top 50 Spotify Songs in 73 Countries (Daily Updated)

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Date: April 2026

1. Introduction

This report documents the data quality assessment and cleaning process performed on the 'Top Spotify Songs in 73 Countries (Daily Updated)' dataset as part of the Global Groove project. The dataset captures daily Top 50 chart rankings across more than 70 countries from October 2023 to June 2025, sourced from the Spotify API via Kaggle.

Due to the file size exceeding Microsoft Excel's row limit of 1,048,576 rows, all cleaning was performed in Power BI using Power Query. This is an industry-standard approach for datasets of this scale and does not affect the validity of the cleaning steps performed.

This dataset will ultimately be merged with a second dataset (Kaggle Historical Spotify Charts, 2017-2021) into a unified Microsoft SQL Server database. The cleaning steps documented here prepare Dataset 2 for that ingestion process.

2. Data Quality Issues Identified

The following issues were identified upon initial review of the raw CSV file in Power Query:

Issue Type	Column(s) Affected	Finding	Cleaning Action
Missing Values	country	NULL values present. These represent the Global Top 50 playlist — not truly missing data. The absence of a country code is itself meaningful.	Filled NULL values with the string 'GLOBAL' to preserve these rows as a valid, queryable category rather than discarding them.
Missing Values	daily_movement, weekly_movement	NULL values present on a track's first appearance in a country. Expected behavior — no prior data exists to compute movement from on day one.	Retained NULLs. Documented as an expected condition, not a data error. Filling with zero would be factually incorrect.
Missing Values	album_name	Some tracks (typically standalone singles) have no associated album entry.	Retained NULLs. Column is designated optional in the database schema.

Issue Type	Column(s) Affected	Finding	Cleaning Action
Incorrect Data Type	snapshot_date	Column stored as plain text string in raw CSV despite containing date values.	Converted to Date type in Power Query. No errors encountered — date formatting was consistent throughout the dataset.
Incorrect Data Type	album_release_date	Column stored as plain text string. Mixed formats were anticipated (full date vs. year-only values).	Converted to Date type in Power Query. No errors encountered — Power BI resolved all values cleanly without requiring manual intervention.
Packed / Delimited Data	artists	Multiple artists stored in a single comma-space delimited string (e.g. 'Lady Gaga, Bruno Mars'). This violates first normal form and prevents artist-level querying.	Split on delimiter ', ' using Power Query Split Column. Produced 7 artist columns, indicating some tracks have up to 7 credited artists. Columns renamed to artist_primary through artist_7. Full relational split into ArtistSong join table will be completed at SQL ingestion.
Duplicate Rows	spotify_id + country + snapshot_date	Potential duplicate entries where the same track appears more than once for the same country on the same date.	Removed duplicates using the composite key: spotify_id + country + snapshot_date. This combination uniquely identifies one chart event. Cross-country and cross-date appearances of the same track are valid and were preserved.

3. Cleaning Steps Performed

In Power BI — Power Query:

1. Imported raw CSV via Get Data > Text/CSV. Selected 'Transform Data' to open Power Query before loading, preventing uncleaned data from entering the report.
2. Replaced NULL values in the 'country' column with the string 'GLOBAL' using Replace Values.
3. Converted 'snapshot_date' from Text to Date type. No errors encountered.
4. Converted 'album_release_date' from Text to Date type. No errors encountered — all values resolved cleanly.

5. Removed duplicate rows on the composite key: spotify_id + country + snapshot_date.
6. Split the 'artists' column by the delimiter ', ' (comma-space) at each occurrence, producing 7 positional artist columns.
7. Renamed resulting columns to: artist_primary, artist_2, artist_3, artist_4, artist_5, artist_6, artist_7. The first position is designated primary as it consistently represents the lead artist in the source data.
8. Retained all audio feature columns (danceability, energy, key, loudness, mode, speechiness, acousticness, instrumentalness, liveness, valence, tempo, time_signature). These are out of scope for the core Discovery Gap analysis but are preserved for future project use and will be loaded into a separate SongAudioFeatures table in the database.
9. Applied 'Close & Apply' to load the fully cleaned dataset into the Power BI report canvas.

4. Columns Added During SQL Ingestion

The following columns do not exist in the raw source data and will be added programmatically during the BULK INSERT process into Microsoft SQL Server. They are documented here for full traceability between the cleaning phase and the database schema.

Column	Hardcoded Value	Reason
source	'Spotify_73Countries'	This dataset will be merged with a second dataset (Kaggle Historical Charts) into a shared ChartEntry table in SQL Server. The source column provides row-level traceability back to the originating dataset, which is critical for debugging ingestion issues and isolating data by source during analysis.
chart_type_id	FK to 'Top 50' in ChartType lookup table	This dataset contains exclusively Top 50 chart data — there is no chart type column in the raw file because only one chart type is present. The chart_type_id will be assigned as a constant during ingestion and will reference the ChartType lookup table. Dataset 1 (Kaggle Historical) includes its own chart column with multiple types and will be handled separately.

5. Remaining Data Limitations

- Artist normalization is partially complete. The 'artists' column has been split into positional columns in Power BI, but the full relational split — extracting unique artist names, assigning integer IDs, and populating an ArtistSong join table — will be completed during SQL ingestion.

Name-level deduplication (e.g. inconsistent casing, encoding differences) is also deferred to the SQL layer.

- Song-level deduplication across both datasets is deferred to SQL ingestion. The same track may appear in both Dataset 1 and Dataset 2. The `spotify_id` field will serve as the merge key to ensure each unique track resolves to one row in the Song table.
- The 'popularity' score is a proprietary Spotify metric with an undisclosed methodology. It cannot be directly compared to stream count data from Dataset 1, which measures popularity by raw stream numbers.
- There is a temporal gap between Dataset 1 (ends December 2021) and Dataset 2 (starts October 2023). Trend analysis spanning this 22-month gap is not possible with the current data sources and should be noted in any time-series visualizations.
- NULL values in 'daily_movement' and 'weekly_movement' are retained as NULL rather than filled with zero. A zero value would imply no change in rank, which is factually different from no prior data existing.
- Year-only values in 'album_release_date' were anticipated based on dataset documentation but were not observed during cleaning. Power BI resolved all date values without errors. If any exist in unreviewed rows, they will display as January 1st of the given year after type conversion.